Visible potentials and one Potts state

All ten MNIST digits | 1,536-state global Potts unit + 784 per-pixel Ising units

On-site energy: $E_i^{\mathrm{on}} = k_i x_i^2/2 + u_i x_i^4/4 + \ell_i x_i$

Bare coefficients are shown; k is not the full effective local curvature.

$s = 0.00$	$s = 1.26$	$s = 2.53$	$s = 3.79$	$s = 5.00$
$t = 5.00$	$t = 3.74$	$t = 2.47$	$t = 1.21$	$t = 0.00$

a

Visible springs

$\sqrt{\langle c_{ij}^2 \rangle_j}$

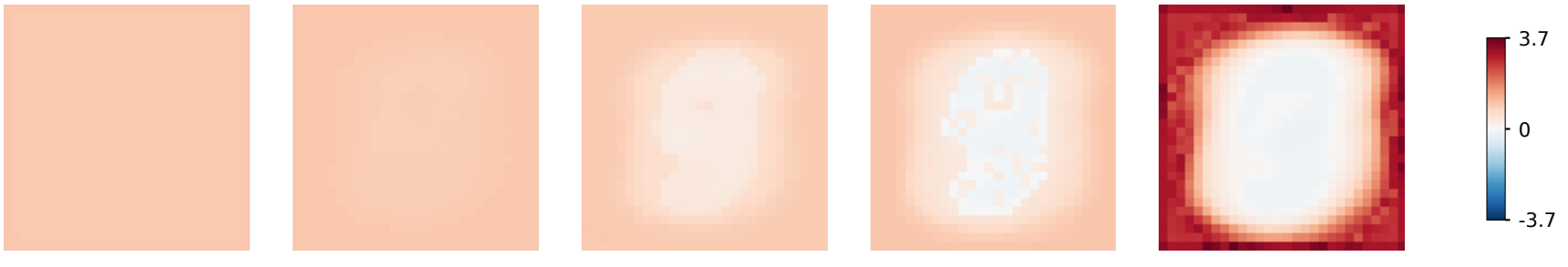


At each pixel: RMS of its incident springs across the full lattice; magnitude only.

b

Bare linear stiffness

$k_i(t)$

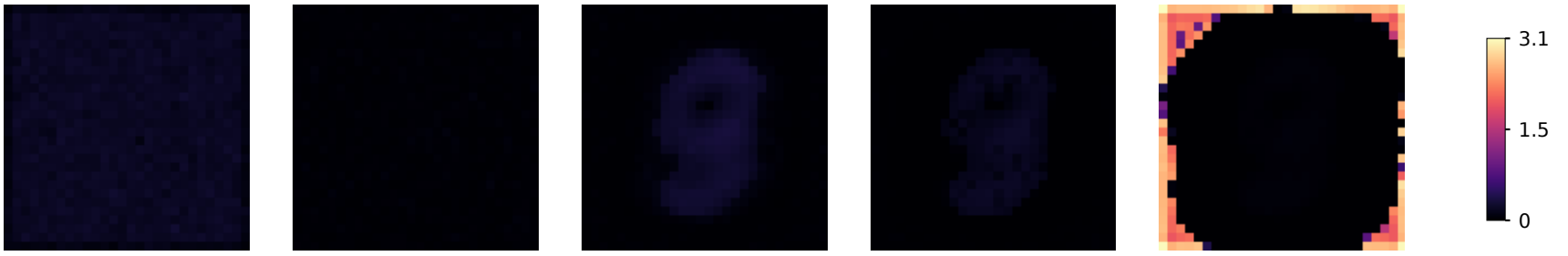


Coefficient of x²/2; signed scale shared across all five times.

c

Duffing coefficient

$u_i(t)$



Coefficient of x⁴/4; this quartic run has no sextic on-site term.

d

Potts state 0 Linear

$W_{0i}(t)$

initial class: 0



W uses the same colour scale for every displayed state on both pages.

e

Potts state 0 Quadratic

$Q_{0i}(t)$

initial class: 0



Q uses the same colour scale for every displayed state on both pages.

SAMPLE TRAJECTORY — PLACEHOLDER

Intermediate trajectory snapshots were not saved; no trajectory was reconstructed.

Linear colour scales; limits are shared across times and across states within each coefficient family.

Generation time s = 5 − t. Saved coefficients, not optimizer history. | Page 1

Three more fixed Potts states

States j = 1, 2, 4 | Shared linear W and Q colour scales across both pages

State preference: $a_j(\mathbf{x}) = b_j + \sum_i W_{ji} x_i + \sum_i Q_{ji} x_i^2$

Initial class denotes the initialization cluster's majority label, not a measured final class.

$s = 0.00$	$s = 1.26$	$s = 2.53$	$s = 3.79$	$s = 5.00$
$t = 5.00$	$t = 3.74$	$t = 2.47$	$t = 1.21$	$t = 0.00$

a

Potts state 1
Linear

$W_{1i}(t)$

initial class: 6



b

Potts state 1
Quadratic

$Q_{1i}(t)$

initial class: 6



Same state across time; shared W and Q scales across both pages.

c

Potts state 2
Linear

$W_{2i}(t)$

initial class: 2



d

Potts state 2
Quadratic

$Q_{2i}(t)$

initial class: 2



Same state across time; shared W and Q scales across both pages.

e

Potts state 4
Linear

$W_{4i}(t)$

initial class: 9



f

Potts state 4
Quadratic

$Q_{4i}(t)$

initial class: 9



Same state across time; shared W and Q scales across both pages.

Linear colour scales; limits are shared across times and across states within each coefficient family.

Generation time s = 5 − t. Saved coefficients, not optimizer history. | Page 2